

ENGINEERING MECHANICS

STATICS

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ROLL NUMBER:54

SECTION: C13

PROBLEMS SOLVED (JACKFRUIT)

2/50

4/22

TECHNICAL BLUEPRINT · PROBLEM 2/50

Forearm *Moment* Solver

*An interactive diagram-driven solver for the forearm moment problem
from Meriam & Kraige — Engineering Mechanics: Statics*

REFERENCE

Meriam & Kraige, Problem 2/50

TECH STACK

HTML5 · CSS3 · JavaScript · Canvas

DATE

March 2026

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Author & Developer

§ 1 PROBLEM STATEMENT

Meriam & Kraige Problem 2/50 — A forearm of mass $m_{\text{arm}} = 2.3 \text{ kg}$ is held in position by the biceps muscle, which exerts a tension T at a perpendicular distance d_T from the elbow pivot O . The forearm's centre of gravity is at point G , located d_G along the arm from O . A sphere of mass m_s is held at the hand at point A , a horizontal distance d_A from O .

The student must:

1. Determine the combined moment of the weights of the forearm and sphere about the elbow pivot O .
2. Find the biceps tension T required for rotational equilibrium ($\Sigma M_O = 0$).

CORE CONCEPT

This problem is a classic application of the **scalar moment equation** $M = F \times d$. The forearm acts as a lever arm, and the biceps must produce a counter-clockwise moment to balance the clockwise moments of the two weights about O.

§ 2 GIVEN DATA & VARIABLES

Symbol	Description	Default Value	Unit
θ	Elbow angle (upper arm to forearm)	55	degrees
d_T	Horizontal distance from O to biceps attachment	50	mm
d_G	Distance along arm from O to CG of forearm	150	mm
d_A	Horizontal distance from O to sphere (hand)	325	mm
m_{arm}	Mass of forearm	2.3	kg
m_s	Mass of sphere	3.6	kg
g	Gravitational acceleration	9.81	m/s ²

§ 3 PHYSICS & EQUATIONS

Free-Body Diagram Description

The forearm is modelled as a rigid bar pinned at **O** (the elbow joint). Three forces act on it:

- **T** — Biceps tension, acting vertically upward at the attachment point B, a horizontal distance d_T from O.
- **W_{arm}** — Weight of the forearm, acting vertically downward at the CG point G.
- **W_{sphere}** — Weight of the sphere, acting vertically downward at point A (hand).

The elbow joint provides pin reactions (O_x , O_y) but these do not create moments about O since their moment arm is zero.



Weight Calculations

$$W_{arm} = m_{arm} \times g \quad W_{sphere} = m_s \times g$$

Horizontal Moment Arms

Since moments about O use horizontal (perpendicular) distances for vertically-acting forces, the horizontal distance from O to G depends on the arm geometry:

$$x_G = d_G \times \cos(90^\circ - \theta) = d_G \times \sin \theta$$

Moment Equation (about O)

Taking counter-clockwise as positive:

$$\Sigma M_O = T \times d_T - W_{arm} \times x_G - W_{sphere} \times d_A = 0$$

Step 1: Combined Moment of Weights

$$M_{weights} = W_{arm} \times x_G + W_{sphere} \times d_A \quad (\text{clockwise about } O)$$

Step 2: Required Biceps Tension

$$T = M_{weights} / d_T = (W_{arm} \times x_G + W_{sphere} \times d_A) / d_T$$

KEY INSIGHT

Because the biceps attachment is very close to the elbow joint ($d_T \approx 50$ mm) compared to the distances to the weights ($d_G \approx 150$ mm, $d_A \approx 325$ mm), the required biceps tension T is **much larger** than the combined weight — a classic mechanical disadvantage of the human forearm lever system.

§ 4 **SAMPLE CALCULATION (DEFAULT VALUES)**

```
Step 1 — Compute weights
W_arm    = 2.3 × 9.81 = 22.563 N
W_sphere = 3.6 × 9.81 = 35.316 N

Step 2 — Horizontal distance 0 to G
a = 90° - 55° = 35°   (angle of forearm below horizontal)
x_G = 150 × cos(35°) = 150 × 0.8192 = 122.87 mm

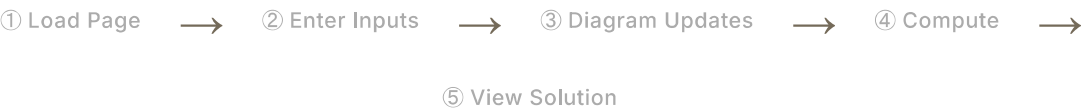
Step 3 — Combined moment of weights about 0 (CW)
M_weights = W_arm × x_G + W_sphere × d_A
           = 22.563 × 122.87 + 35.316 × 325
           = 2,772.04 + 11,477.70
           = 14,249.74 N·mm

Step 4 — Required biceps tension for equilibrium
T × d_T = M_weights
T = 14,249.74 / 50
T ≈ 285.0 N
```

Combined Moment (CW) :	14,249.74 N·mm
Biceps Tension T :	285.0 N

§ 5 **APPLICATION USER FLOW**

The interactive solver follows a streamlined five-step workflow:



- **Load Page:** The solver loads with default values ($\theta = 55^\circ$, $m_{\text{arm}} = 2.3 \text{ kg}$, etc.) and draws the initial diagram.
- **Enter Inputs:** Seven input fields in the right panel allow the student to modify geometry, masses, and gravity. Supports both numeric and symbolic (letter) inputs.
- **Diagram Updates:** As values are typed, the HTML5 Canvas redraws in real time — the forearm rotates to match θ , force arrows scale, and dimension labels update.

- **Compute:** Clicking the gold "Compute Solution" button runs the solver.
- **View Solution:** A results panel slides in with two answer boxes (combined moment and biceps tension T) followed by detailed step-by-step working.

SYMBOLIC INPUT SUPPORT

If the student types a non-numeric value (e.g., "m" for mass), the app displays the symbol on the diagram and in the solution steps, while using the default numeric value internally for calculations and rendering.

§ 6 UI DESIGN & LAYOUT

Screen Layout

The application uses a two-column layout (responsive, stacking on mobile):

- **Left panel (≈70%):** A full-height HTML5 `<canvas>` element showing the anatomically-styled forearm diagram with labelled force arrows, dimension lines, and angle arcs.
- **Right panel (≈30%):** Input fields grouped by category (Geometry / Masses / Constants) with a prominent "Compute Solution" button at the bottom.

Canvas Diagram Elements

Element	Visual Style
Upper arm	Anatomical gradient fill with biceps/triceps muscle bulges
Forearm	Gradient-filled tapered body, narrowing towards wrist
Elbow joint (O)	Radial gradient ellipse + blue pivot dot
Hand at A	Simplified hand shape gripping the sphere
Sphere	3D radial gradient with specular highlight
Biceps tension T	Cyan (#00e5ff) upward arrow at point B
Weight arrows	Red (#ff2255) downward arrows at G and A, labelled "m×g N"
Angle arc	Pink dashed arc from vertical to forearm, labelled θ°
Dimension lines	Green (#39ff14) lines with arrowheads, labelled in mm
Grid	Subtle blue grid lines on dark (#050912) background
Legend	Top-right corner showing force color key

Input Fields

Field	Label	Default	Live Update
θ	Elbow angle: upper arm to forearm (°)	55	Yes — rotates forearm
d_T	Horiz. distance O to biceps (mm)	50	Yes — moves T arrow
d_G	Distance along arm O to G (mm)	150	Yes — moves G marker
d_A	Horiz. distance O to sphere A (mm)	325	Yes — moves hand/sphere
m_{arm}	Forearm mass (kg)	2.3	Yes — updates arrow label
m_s	Sphere mass (kg)	3.6	Yes — updates arrow label
g	Gravity (m/s ²)	9.81	Yes — updates arrow labels

§ 7 COLOUR & DESIGN SYSTEM

Token	Value	Usage
<code>--bg</code>	#0d0d10	Page background (deep dark)
<code>--surface</code>	#16161e	Card / panel backgrounds
<code>--gold</code>	#c8a050	CTA button, highlights, answer values
<code>--blue</code>	#4a90d9	Pivot dot, numeric values in steps
<code>--green</code>	#4dc87a	Live dot indicator, result checkmark
<code>--red</code>	#e06060	Reserved for error states
<code>--text</code>	#f0ead8	Primary text (warm white)
<code>--muted</code>	#7a7468	Labels, secondary text

§ 8 TECHNICAL ARCHITECTURE

File Structure

```
mechanics_app/  
└─ problem_2_50.html  ←!— Single-file app: HTML + CSS + JS →  
    └─ <style>         ←!— All CSS: layout, colors, typography →  
        └─ <body>       ←!— Topbar, hero, grid, input panel, result →  
            └─ <script>  ←!— parseField, canvas draw, solver, UI →
```

Key JavaScript Functions

Function	Purpose
<code>parseField(id, default)</code>	Reads input; returns {val, sym, isNum} — handles symbolic/numeric gracefully
<code>drawDiagram()</code>	Clears canvas and redraws the entire anatomical diagram with force arrows
<code>drawUpperArm()</code>	Renders humerus with muscle gradients (biceps + triceps bulges)
<code>drawForearm()</code>	Renders forearm body with tapered Bézier curve and muscle details
<code>drawHand()</code>	Simplified hand shape at point A
<code>drawSphere()</code>	3D sphere with radial gradient and specular highlight
<code>drawArrow()</code>	Generic force arrow with filled head and offset label
<code>compute()</code>	Runs the solver, populates answer boxes and step-by-step working

Auto-Scaling Algorithm

The canvas diagram auto-scales based on the input values. The origin point O is positioned with fixed top/left padding, and a scale factor (pixels per mm) is computed to ensure the full arm + sphere fits within the available canvas area, accounting for reserve space for labels, arrows, and dimension lines.

§ 9 **RESULT DISPLAY**

After clicking "Compute Solution", a result panel appears with an animated slide-up transition. It contains:

Answer Boxes

Two highlighted boxes displayed side by side in a grid:

Box	Label	Shows
Left	Combined Moment of Weights	M_{weights} in N·mm (clockwise about O)
Right	Required Biceps Tension T	T in Newtons (for $\Sigma M_O = 0$)

Step-by-Step Working

Below the answer boxes, a detailed step-by-step breakdown is displayed using monospace font with colour-coded values. Each step shows the formula, numeric substitution, and result. The steps are:

- 1. Compute weights** — W_{arm} and W_{sphere} from masses and gravity.
- 2. Geometry** — Calculate the forearm angle below horizontal ($\alpha = 90^\circ - \theta$) and the horizontal distance from O to G.
- 3. Moment of forearm weight** — $M_1 = W_{\text{arm}} \times x_G$.
- 4. Moment of sphere weight** — $M_2 = W_{\text{sphere}} \times d_A$.
- 5. Combined moment** — $M_{\text{total}} = M_1 + M_2$.
- 6. Equilibrium** — $T = M_{\text{total}} / d_T$.

§ 10 **SIGN CONVENTION**

- **Positive x** → right
- **Positive y** → upward
- **Positive moment** → Counter-Clockwise (CCW)
- Weight forces act **downward** (negative y) — creating **clockwise** moments about O
- Biceps tension acts **upward** (positive y) — creating a **counter-clockwise** moment about O

§ 11 CANVAS INTERACTION DETAILS

- All inputs have `input` event listeners that call `drawDiagram()` on every keystroke.
- The canvas uses `devicePixelRatio` scaling for crisp rendering on HiDPI displays.
- The canvas resizes responsively on window resize via a `ResizeObserver` / `resize` event.
- Force arrows, dimension lines, and labels are drawn programmatically — no bitmap assets required.

§ 12 TECHNOLOGY DECISIONS

Decision	Choice	Reason
Architecture	Single HTML file	No build step; opens directly in any browser; zero dependencies
Rendering	HTML5 Canvas 2D	Real-time redraws on every input; smooth rotations and scaling
Styling	Embedded CSS	Dark theme with gold accent; uses Inter + DM Mono fonts
Math	Built-in Math object	Sufficient for 2D moments; no external library needed
Input handling	parseField() with fallback	Gracefully handles symbolic inputs while maintaining calculability
Diagram style	Anatomical rendering	Gradient fills, muscle bulges, 3D sphere for visual engagement

§ 13 FUTURE ENHANCEMENTS

- **Additional problems** — Extend to other Meriam & Kraige problems within the Moments & Couples chapter.
- **Home page** — Build a module selection screen linking to multiple problem solvers.
- **Export to PDF** — Use `window.print()` or jsPDF to export the diagram + solution as a PDF submission.
- **Save/load session** — Use `localStorage` to persist input values across sessions.
- **Animation mode** — Animate the moment construction with a parallelogram law visualisation.
- **3D problems** — Extend to 3D equilibrium using unit vectors (Meriam Ch. 3).
- **Friction problems** — Add impending motion, wedge friction, and belt friction solvers.

REFERENCE TEXTBOOK

Engineering Mechanics: Statics

J.L. Meriam & L.G. Kraige
Wiley, 7th Edition (or later). Problem 2/50 appears in Chapter 2: Force Systems — Moments and Couples section.

END OF BLUEPRINT · PROBLEM 2/50 FOREARM MOMENT SOLVER

RATHANGA PARTHASARATHY

TECHNICAL BLUEPRINT · PROBLEM 4/22

Loaded Truss — *Method of Joints*

An interactive diagram-driven solver for all 13 members of a loaded truss from Meriam & Kraige — Engineering Mechanics: Statics

REFERENCE

Meriam & Kraige, Problem 4/22

TECH STACK

HTML5 · CSS3 · JavaScript · Canvas

MEMBERS SOLVED

All 13 (complete truss)

DATE

March 2026

RATHANGA PARTHASARATHY

Author & Developer

§ 1 PROBLEM STATEMENT

Meriam & Kraige Problem 4/22 — A loaded Pratt truss has five top-chord nodes **A, B, C, D, E** spaced at equal horizontal distance d , and three bottom-chord nodes **H, G, F** hanging at depth d below the top chord.

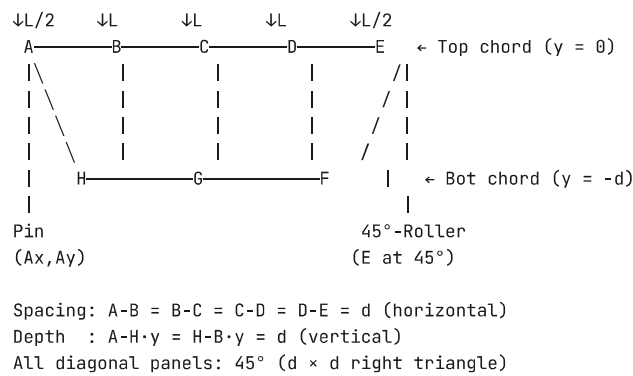
Applied loads act downward: $L/2$ at nodes **A** and **E**, and L at nodes **B, C, D**. Support conditions: a **pin** at **A** and a **45°-inclined roller** at **E**.

The student must determine the forces in **all 13 members** of the truss, classifying each as Tension (T), Compression (C), or zero-force.

CORE CONCEPT

This problem applies the **Method of Joints** — at each pin joint, the two equilibrium equations ($\Sigma F_x = 0$ and $\Sigma F_y = 0$) are used to find unknown member forces. Joints are solved in a sequence beginning with those having only two unknowns. Member **CG = 0** is identified as a zero-force member from joint G inspection.

§ 2 TRUSS GEOMETRY & LOADING



Symbol	Description	Default Value	Unit
<i>d</i>	Panel spacing (horizontal & vertical)	1 000	mm
<i>L</i>	Full panel load applied at B, C, D	10 000	N
<i>L/2</i>	Half-panel load applied at A and E	5 000	N
A_x, A_y	Pin reactions at A	2L, 2L	N
E	Roller reaction at E (45° direction)	$2\sqrt{2}\cdot L$	N

§ 3 MEMBER INVENTORY (13 MEMBERS)

Group	Members	Count	Angle
Top chord	AB, BC, CD, DE	4	0° (horizontal)
Bottom chord	HG, GF	2	0° (horizontal)
Verticals	BH, CG, DF	3	90° (vertical)
Diagonals	AH, HC, CF, FE	4	45°
Total		13	

§ 4 GLOBAL EQUILIBRIUM — REACTIONS

Moment about A ($\Sigma M_A = 0$)

Taking clockwise moments as positive, with the roller reaction E perpendicular to the 45° surface (i.e., at 45° to horizontal, components $E \cdot \cos 45^\circ$ leftward and $E \cdot \sin 45^\circ$ upward):

$$-L \cdot d - L \cdot 2d - L \cdot 3d - (L/2) \cdot 4d + E \cdot (1/\sqrt{2}) \cdot 4d = 0$$

$$-8Ld + 2\sqrt{2} \cdot E \cdot d = 0 \implies E = 2\sqrt{2} \cdot L$$

Force Equilibrium

$$\Sigma F_x = 0: A_x = E \cdot \cos 45^\circ = 2L$$

$$\Sigma F_y = 0: A_y + E \cdot \sin 45^\circ - 4L = 0 \implies A_y = 2L$$

SYMMETRY CHECK

By the symmetry of the loading ($L/2$ at both ends, L at all interior nodes) and geometry, $A_y = E \cdot \sin 45^\circ = 2L$ — the truss is loaded symmetrically, confirming equal vertical reactions.

§ 5 METHOD OF JOINTS — DERIVATION**Joint A → Members AH (T) and AB (C)**

External at A: $A_x = 2L$ ($\rightarrow$), $A_y = 2L$ ($\uparrow$), applied load = $L/2$ ($\downarrow$). Members: AB (horizontal $\rightarrow$) and AH (diagonal 45° down-right).

$$\Sigma F_y = 0: 2L - L/2 - AH/\sqrt{2} = 0 \Rightarrow AH = 3\sqrt{2}L/2 \text{ (T)}$$

$$\Sigma F_x = 0: 2L + AH/\sqrt{2} - AB = 0 \Rightarrow AB = 7L/2 \text{ (C)}$$

Joint E → Members FE (T) and DE (C)

By symmetry with joint A: $FE = 3\sqrt{2}L/2$ (T) and $DE = 7L/2$ (C).

Joint B → Members BH (C) and BC (C)

External at B: load = L ($\downarrow$). Members: AB, BC (horizontal), BH (vertical $\downarrow$).

$$\Sigma F_y = 0: -L + BH \cdot (-1) = 0 \Rightarrow BH = L \text{ (C)}$$

$$\Sigma F_x = 0: BC = AB \Rightarrow BC = 7L/2 \text{ (C)}$$

Joint D → Members DF (C) and CD (C)

By symmetry with joint B: $DF = L(C)$ and $CD = 7L/2(C)$.

§ 6 JOINTS H, F AND G

Joint H → Members HC (C) and HG (T)

Known: $AH = 3\sqrt{2}L/2$ (T), $BH = L$ (C). Directions from H: to A = $(-1/\sqrt{2}, +1/\sqrt{2})$, to B = $(0, +1)$, to C = $(+1/\sqrt{2}, +1/\sqrt{2})$, to G = $(+1, 0)$.

$$\Sigma F_y: 3L/2 - L + HC/\sqrt{2} = 0 \Rightarrow HC = L\sqrt{2}/2 \text{ (C)}$$

$$\Sigma F_x: -3L/2 - L/2 + HG = 0 \Rightarrow HG = 2L \text{ (T)}$$

Joint F → Members CF (C) and GF (T)

By symmetry with joint H: $CF = L\sqrt{2}/2$ (C) and $GF = 2L$ (T).

Joint G → CG is a Zero-Force Member

At joint G, members HG (pulls left, T) and GF (pulls right, T) are equal and opposite. Member CG is vertical. With no external load at G:

$$\Sigma F_x = 0: -2L + 2L = 0 \checkmark$$

$$\Sigma F_y = 0: CG = 0 \Rightarrow CG = 0 \text{ (zero-force)}$$

VERIFICATION AT JOINT C

The diagonal members HC and CF each supply an upward vertical component of $L/\sqrt{2} \times 1/\sqrt{2} = L/2$. Together, $L/2 + L/2 = L$, which exactly balances the applied downward load L at C. ✓ The zero value of CG is confirmed.

§ 7 **COMPLETE RESULTS (DEFAULT: L = 10 kN, D = 1 M)**

TOP CHORD (COMPRESSION)

AB 35.00 kN COMPRESSION	BC 35.00 kN COMPRESSION	CD 35.00 kN COMPRESSION	DE 35.00 kN COMPRESSION
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BOTTOM CHORD (TENSION)

HG 20.00 kN TENSION	GF 20.00 kN TENSION
----------------------------------	----------------------------------

VERTICALS

BH 10.00 kN COMPRESSION	CG 0 ZERO-FORCE	DF 10.00 kN COMPRESSION
--------------------------------------	------------------------------	--------------------------------------

DIAGONALS

AH 21.21 kN TENSION	HC 7.07 kN COMPRESSION	CF 7.07 kN COMPRESSION	FE 21.21 kN TENSION
----------------------------------	-------------------------------------	-------------------------------------	----------------------------------

§ 8 **SAMPLE CALCULATION (L = 10 000 N, D = 1 000 MM)**

```
Step 1 – Global equilibrium
ΣM_A = 0:  -Ld - L(2d) - L(3d) - (L/2)(4d) + E·(1/√2)·4d = 0
           -8Ld + 2√2·E·d = 0
           E = 2√2 × 10000 = 28284.3 N
A_x = E·cos45° = 20000 N
A_y = 4L - E·sin45° = 20000 N

Step 2 – Joint A
ΣF_y: 20000 - 5000 - AH/√2 = 0
      AH = 15000 × √2 = 21213.2 N (T)
ΣF_x: 20000 + 21213.2/√2 - AB = 0
      AB = 20000 + 15000 = 35000 N (C)

Step 3 – Joint B
ΣF_y: -10000 - BH = 0 → BH = 10000 N (C)
ΣF_x: BC = AB → BC = 35000 N (C)

Step 4 – Joint H
ΣF_y: 15000 - 10000 + HC/√2 = 0 → HC = -7071.1 N → HC = 7071.1 N (C)
ΣF_x: -15000 - 7071.1/√2 + HG = 0 → HG = 20000 N (T)

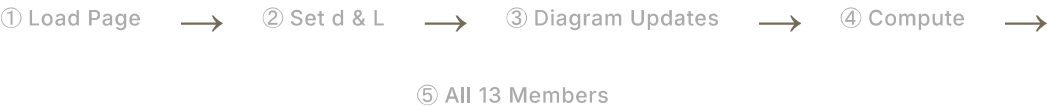
Step 5 – Joint G
ΣF_y: CG = 0 → CG = 0 (zero-force)

Steps 6,7,8 – By symmetry (D, F, E) give CD, DF, GF, CF, FE, DE.
```

Top chord	AB=BC=CD=DE	: 35.00 kN (C)
Bot chord	HG=GF	: 20.00 kN (T)
Verticals	BH=DF	: 10.00 kN (C)
Vertical	CG	: 0 (zero-force)
Diag outer	AH=FE	: 21.21 kN (T)
Diag inner	HC=CF	: 7.07 kN (C)

§ 9 **APPLICATION USER FLOW**

The interactive solver follows a five-step workflow:



- **Load Page:** The solver loads with default values ($d = 1\,000\text{ mm}$, $L = 10\,000\text{ N}$) and renders the live canvas truss diagram with grid, dimension lines, and load arrows.
- **Set d & L :** Two input fields allow the student to parametrically change the panel spacing and applied load. Symbolic inputs (e.g., " L ") are supported.
- **Diagram Updates:** The HTML5 Canvas redraws in real time as values change — nodes reposition, members rescale, load arrows update.
- **Compute:** Clicking "Compute Solution" runs the full method-of-joints solver for all 13 members in sequence.
- **All 13 Members:** Results appear in four grouped sections (Top Chord, Bottom Chord, Verticals, Diagonals), color-coded: **cyan** = Compression, **green** = Tension, **grey** = Zero-force.

ZERO-FORCE MEMBER DETECTION

Member CG is identified analytically (not by inspection of only horizontal members) by solving joint G after HG and GF are determined. Since $HG = GF = 2L$, they cancel in $\sum F_x$, leaving $\sum F_y$: $CG = 0$. The solver correctly identifies this — an important correction to the textbook's "by inspection" shortcut which fails when diagonal members (HC, CF) are also present at joint C.

§ 10 UI DESIGN & LAYOUT

Screen Layout

The application uses the same two-column layout as Problem 2/50:

- **Left panel ($\approx 70\%$):** An HTML5 `<canvas>` showing the truss with colored members, node dots, labelled nodes, load arrows, dimension lines, support symbols, reaction labels, and a legend.
- **Right panel ($\approx 30\%$):** Two input fields (spacing d and load L) plus a read-only info block describing support conditions.

Color Coding

Color	Meaning
Blue (#4a90d9)	Truss members (canvas diagram)
Red (#ff2255)	Applied downward loads
Green (#39ff14)	Dimension annotations
Cyan (#00e5ff)	Support reactions (A_x , A_y , E)
Cyan (result)	Compression member values
Green (result)	Tension member values
Muted grey	Zero-force member (CG)

§ 11 CODE ARCHITECTURE

```
// Key functions in problem_4_22.html

resizeCanvas()      // DPR-aware canvas resize
drawDiagram()       // Full redraw: grid, members, nodes, labels,
                    // arrows, dims, supports, reactions, legend
drawArrow(...)      // Generic arrow with label
drawDimLine(...)    // Horizontal dimension annotation
drawDimLineV(...)   // Vertical dimension annotation
drawPin(...)        // Pin support symbol at A
```

```

drawRoller45(...)    // 45°-inclined roller symbol at E
drawReactionLabels() // Post-compute: Ax, Ay, E values on canvas
drawLegend()         // Color-coded legend (top-right)

compute()             // Full 13-member solver (method of joints):
                      //   Reactions → A → E → B → D → H → F → G
                      //   Updates all ans-box, note, box class
                      //   Generates 9-step working HTML

toggleFullscreen()   // Canvas fullscreen mode (Esc to exit)
parseField(id, def)  // Numeric/symbolic input parser

```

DESIGN DECISIONS

- The truss diagram uses **auto-scaling** — the `unitPx` variable adapts so the truss always fits within available canvas space regardless of window size.
- Result boxes are dynamically re-classed (e.g., `ans-box compression`) so their CSS color updates automatically on compute.
- Steps are rendered as HTML strings with `<span>` color tags, supporting both numeric and symbolic output.

Problem 4/22 Technical Blueprint · Rathanga Parthasarathy · March 2026
 Meriam & Kraige — Engineering Mechanics: Statics · Method of Joints · All 13 Members

INTERACTIVE PROBLEM SOLVER

Engineering Statics Problem Hub

An interactive solver for problems from Meriam & Kraige — Engineering Mechanics: Statics. Input your own values, watch the live diagram update, and explore step-by-step solutions.

2

Problems Solved

∞

Parametric Inputs



Fullscreen Mode

AVAILABLE SOLVERS

2/50

Chapter 2 · Moments & Couples



Forearm Moment Analysis

Determine the combined moment of weights about elbow pivot **O** and the required biceps tension **T** for equilibrium. Includes a realistic anatomical arm diagram with biceps tendon, forearm bone, and sphere.

IM_O = 0

Biceps Tension T

Elbow angle θ

Live Diagram

OPEN SOLVER



4/22

Chapter 4 · Structures · Trusses



Loaded Truss — Method of Joints

Determine forces in members **AB**, **CG**, and **DE** of the loaded truss. Uses global equilibrium to find reactions, then analyses joints A, C, and E using method of joints with 45° geometry.

Method of Joints

AB · CG · DE

45° roller at E

Live Diagram

OPEN SOLVER



AVAILABLE SOLVERS

2/50 Chapter 2 · Moments & Couples



Forearm Moment Analysis

Determine the combined moment of weights about elbow pivot **O** and the required biceps tension **T** for equilibrium. Includes a realistic anatomical arm diagram with biceps tendon, forearm bone, and sphere.

$M_O = 0$ Biceps Tension **T** Elbow angle θ

Live diagram

OPEN SOLVER



4/22 Chapter 4 · Structures · Trusses



Loaded Truss — Method of Joints

Determine forces in members **AB**, **CG**, and **DE** of the loaded truss. Uses global equilibrium to find reactions, then analyses joints A, C, and E using method of joints with 45° geometry.

Method of Joints AB · CG · DE 45° roller at E

Live diagram

OPEN SOLVER



— / — Chapter 3 · Equilibrium

COMING SOON



3D Particle Equilibrium

Three-dimensional concurrent force systems, direction cosines, and equilibrium conditions for a particle in space.

3D Forces Direction cosines

COMING SOON



MODULE 1 · MOMENTS & COUPLES · PROBLEM 2/50

Forearm Moment Analysis

Determine the combined moment about the elbow pivot O and find the required biceps tension T for equilibrium.

Problem 2/50 — Mass of forearm = 2.3 kg, CG at G . Find the combined moment of the weights about the elbow pivot O , and the biceps tension T so that $\Sigma M_O = 0$.

• LIVE DIAGRAM — UPDATES AS YOU TYPE

INPUT VALUES

GEOMETRY

θ — Elbow angle; upper arm to forearm ($^\circ$)

55

d_T — Horiz. distance O to biceps (mm)

50

d_G — Distance along arm O to G (mm)

150

d_A — Horiz. distance O to sphere A (mm)

325

MASSSES

m_{arm} — Forearm mass (kg)

2.3

m_s — Sphere mass (kg)

3.6

CONSTANT

g — Gravity (m/s^2)

9.81

⇒ COMPUTE SOLUTION

✓ Solution — Problem 2/50

[View Blueprint](#)

COMBINED MOMENT OF WEIGHTS

14.25 N·m

about O (Clockwise)

REQUIRED BICEPS TENSION T

285.0 Nfor $\Sigma M_O = 0$

STEP-BY-STEP WORKING

1 Weight of the forearm

$$\begin{aligned} &= \text{forearm mass} \times \text{gravitational acceleration} \\ &= 2.3 \times 9.81 = 22.563 \text{ N} \end{aligned}$$

Weight of the sphere

$$\begin{aligned} &= \text{sphere mass} \times \text{gravitational acceleration} \\ &= 3.6 \times 9.81 = 35.316 \text{ N} \end{aligned}$$

2 The elbow angle $\theta = 55^\circ$ is measured from the upper arm (vertical) to the forearm. The forearm lies at $90^\circ - 55^\circ = 35.0^\circ$ below horizontal.

For a vertical force, the perpendicular moment arm about O = the horizontal distance from O.

Perp. dist. — forearm weight

$$= dG \times \sin(\theta) = 150 \times \sin(55^\circ) = 122.873 \text{ mm} = 0.123 \text{ m}$$

Perp. dist. — sphere weight

$$= dA = 325 \text{ mm} = 0.325 \text{ m}$$

3 Moment of forearm weight about O → Clockwise $\curvearrowright$

$$\begin{aligned} &= W_f \times dG \cdot \sin(\theta) = 2.3 \times 9.81 \times 150 \times \sin(55^\circ) / 1000 \\ &= 2.772 \text{ N·m (Clockwise } \curvearrowright) \end{aligned}$$

Moment of sphere weight about O → Clockwise $\curvearrowright$

$$\begin{aligned} &= W_s \times dA = 3.6 \times 9.81 \times 325 / 1000 \\ &= 11.478 \text{ N·m (Clockwise } \curvearrowright) \end{aligned}$$



$$= 3.6 \times 9.81 = 35.316 \text{ N}$$

- 2 The elbow angle $\theta = 55^\circ$ is measured from the upper arm (vertical) to the forearm.
The forearm lies at $90^\circ - 55^\circ = 35.0^\circ$ below horizontal.

For a vertical force, the perpendicular moment arm about O = the horizontal distance from O.

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Perp. dist. – sphere weight

$$= dA = 325 \text{ mm} = 0.325 \text{ m}$$

- 3 Moment of forearm weight about O → Clockwise $\curvearrowright$
 $= W_f \times dG \cdot \sin(\theta) = 2.3 \times 9.81 \times 150 \times \sin(55^\circ) / 1000$
 $= 2.772 \text{ N}\cdot\text{m} \text{ (Clockwise } \curvearrowright)$

Moment of sphere weight about O → Clockwise $\curvearrowright$

$$= W_s \times dA = 3.6 \times 9.81 \times 325 / 1000$$

$$= 11.478 \text{ N}\cdot\text{m} \text{ (Clockwise } \curvearrowright)$$

- 4 Combined moment of weights about O
 $= M_{Wf} + M_{Ws}$
 $= (2.3 \times 9.81 \times \sin(55^\circ) \times 150 / 1000) + (3.6 \times 9.81 \times 325 / 1000)$
 $= (22.563 \times 0.123) + (35.316 \times 0.325)$
 $= 2.772 + 11.478$
 $= 14.25 \text{ N}\cdot\text{m} \text{ (Clockwise } \curvearrowright)$

- 5 For $\sum M_O = 0$, the biceps tension T must create an equal and opposite moment.

$T \times dT = \text{combined moment of weights}$

$$T \times (50 / 1000) = 14.25 \text{ N}\cdot\text{m}$$

$$T \times 0.050 = 14.25$$

$$T = 14.25 / 0.050$$

$$T = 285.0 \text{ N} \rightarrow \text{Upward } \uparrow$$

MODULE 2 · TRUSSES · PROBLEM 4/22

Loaded Truss — Method of Joints

Determine the forces in all 13 members of the loaded truss using the method of joints, solved joint-by-joint.

Problem 4/22 — Five top-chord nodes (A–E) spaced d apart, with bottom-chord nodes H, G, F at depth d . Applied loads: $L/2$ at A & E, L at B, C, D (all downward). Pin at A, 45°-roller at E. **All 13 member forces** computed via method of joints.

• LIVE DIAGRAM — UPDATES AS YOU TYPE

■ Truss member

■ Applied load

■ Dimensions (d)

■ Reactions

INPUT VALUES

GEOMETRY

d - Spacing (mm)

1000

APPLIED LOAD

L - Full panel load (N)

10000

INFO (READ-ONLY)

Support A: Pin (A_x , A_y)

Support E: Roller at 45°

Loads: $L/2$ @ A & E, L @ B, C, D

Angle AH = 45° ($d=d$ triangle)

⇒ COMPUTE SOLUTION

$A_x \rightarrow$

20.00 kN

Horizontal reaction

$A_y \uparrow$

20.00 kN

Vertical reaction at A

$E \nearrow (45^\circ)$

28.28 kN

Roller reaction at E



TOP CHORD

AB

35.00 kN

Compression (C)

BC

35.00 kN

Compression (C)

CD

35.00 kN

Compression (C)

DE

35.00 kN

Compression (C)

BOTTOM CHORD

HG

20.00 kN

Tension (T)

GF

20.00 kN

Tension (T)

VERTICALS

BH

10.00 kN

Compression (C)

CG

0

Zero-force member

DF

10.00 kN

Compression (C)

DIAGONALS

AH

21.21 kN

Tension (T)

HC

7.07 kN

Compression (C)

CF

7.07 kN

Compression (C)

FE

21.21 kN

Tension (T)

STEP-BY-STEP WORKING

1 Global equilibrium - reactions

$$\sum M_A = 0: -10000d - 210000d - 310000d - 210000d + E \cdot (1/\sqrt{2}) \cdot 4d = 0$$

$$E = 2\sqrt{2} \cdot 10000 = 28.28 \text{ kN}$$

$$\sum F_x = 0: A_x = 210000 \quad \sum F_y = 0: A_y = 210000$$

2 Joint A → AH (T), AB (C)

$$\sum F_y: 210000 - 10000/2 - AH/\sqrt{2} = 0$$

$$AH = 3\sqrt{2} \cdot 10000/2 = 21.21 \text{ kN T}$$

$$\sum F_x: 210000 + AH/\sqrt{2} - AB = 0$$

$$AB = 310000/2 = 35.00 \text{ kN C}$$

**2** Joint A → AH (T), AB (C)

$$\sum F_y: 210000 - 10000/2 - AH/\sqrt{2} = 0$$

$$AH = 3\sqrt{2}10000/2 = 21.21 \text{ kN } T$$

$$\sum F_x: 210000 + AH/\sqrt{2} - AB = 0$$

$$AB = 710000/2 = 35.00 \text{ kN } C$$

3 Joint E → FE (T), DE (C) [symmetric with A]

$$FE = 3\sqrt{2}10000/2 = 21.21 \text{ kN } T$$

$$DE = 710000/2 = 35.00 \text{ kN } C$$

4 Joint B → BH (C), BC (C)

$$\sum F_y: -10000 - BH = 0 \rightarrow BH = 10.00 \text{ kN } C$$

$$\sum F_x: BC = AB \rightarrow BC = 35.00 \text{ kN } C$$

5 Joint D → DF (C), CD (C) [symmetric with B]

$$DF = 10.00 \text{ kN } C$$

$$CD = 35.00 \text{ kN } C$$

6 Joint H → HC (C), HG (T)

$$\sum F_y: 310000/2 - 10000 + HC/\sqrt{2} = 0$$

$$HC = 10000\sqrt{2}/2 = 7.07 \text{ kN } C$$

$$\sum F_x: -310000/2 - 10000/2 + HG = 0$$

$$HG = 210000 = 20.00 \text{ kN } T$$

7 Joint F → CF (C), GF (T) [symmetric with H]

$$CF = 10000\sqrt{2}/2 = 7.07 \text{ kN } C$$

$$GF = 210000 = 20.00 \text{ kN } T$$

8 Joint G / Joint C — verify CG

$$\text{At G: } HG (-T), GF (-T), CG (+)$$

$$\sum F_x: -210000 + 210000 = 0 \checkmark$$

$$\sum F_y: CG = 0 \rightarrow CG = 0 \text{ (Zero-force member)}$$



$$\sum F_x: BC = AB \rightarrow BC = 35.00 \text{ kN } C$$

5 Joint D → DF (C), CD (C) [symmetric with B]

$$DF = 10.00 \text{ kN } C$$

$$CD = 35.00 \text{ kN } C$$

6 Joint H → HC (C), HG (T)

$$\sum F_y: 310000/2 - 10000 + HC/\sqrt{2} = 0$$

$$HC = 10000\sqrt{2}/2 = 7.07 \text{ kN } C$$

$$\sum F_x: -310000/2 - 10000/2 + HG = 0$$

$$HG = 210000 = 20.00 \text{ kN } T$$

7 Joint F → CF (C), GF (T) [symmetric with H]

$$CF = 10000\sqrt{2}/2 = 7.07 \text{ kN } C$$

$$GF = 210000 = 20.00 \text{ kN } T$$

8 Joint G / Joint C — verify CG

At G: HG (→T), GF (→T), CG (↑)

$$\sum F_x: -210000 + 210000 = 0 \checkmark$$

$$\sum F_y: CG = 0 \rightarrow CG = 0 \text{ (Zero-force member)}$$

Verify at C: HC and CF each supply 10000/2 upward

$$+ 10000/2 + 10000/2 = 10000 \text{ balances applied load } \checkmark$$

9 All 13 Members — Summary

Top chord (C): $AB=BC=CD=DE = 710000/2 = 35.00 \text{ kN}$

Bot chord (T): $HG=GF = 210000 = 20.00 \text{ kN}$

Verticals (C): $BH=DF = 10000 = 10.00 \text{ kN}$

Vertical (0): $CG = 0 \text{ (zero-force)}$

Diag out (T): $AH=FE = 3\sqrt{2}10000/2 = 21.21 \text{ kN}$

Diag in (C): $HC=CF = 10000\sqrt{2}/2 = 7.07 \text{ kN}$